EE559 (EE 559) Course Project Proposal

Adaptive Fraud Detection with Dynamic Thresholding and Cost-Sensitive Learning

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1. Project Objective and Motivation

The goal of this project is to build a supervised machine learning pipeline for credit card fraud detection and extend it with a more realistic decision-making layer based on dynamic thresholding and cost-sensitive analysis. While fraud detection is commonly treated as a standard binary classification problem, real-world fraud systems must make decisions under extreme class imbalance and unequal error costs. Missing a fraudulent transaction can be financially expensive, while flagging too many legitimate transactions can reduce user trust and increase operational burden.

This project will therefore study not only predictive performance, but also how decision thresholds and model behavior change under different fraud-cost assumptions. The final system will compare multiple supervised learning models and analyze which model provides the best trade-off between recall, precision, and operational usability.

2. Dataset Details

The proposed dataset is the **Kaggle Credit Card Fraud Detection Dataset**, a widely used benchmark for imbalanced binary classification.

- **Task Type:** Binary classification
- **Prediction Target:** Fraudulent transaction (1) vs. legitimate transaction (0)
- **Number of Samples:** 284,807 transactions
- **Number of Features:** 30 total input variables
- **Feature Description:**
  - 28 anonymized PCA-transformed features: `V1` to `V28`
  - `Time`
  - `Amount`
- **Class Distribution:** Highly imbalanced; only 492 fraud cases (approximately 0.17%)

This dataset is appropriate for the course because it supports supervised learning, preprocessing, model comparison, class imbalance handling, regularization, and strong experimental analysis.

3. Problem Formulation

This project is formulated as a **supervised binary classification problem**. Given a transaction feature vector $\mathbf{x} \in \mathbb{R}^d$, the model predicts whether the transaction is fraudulent:

$$f(\mathbf{x}) \to y, \quad y \in \{0, 1\}.$$

Here, $y = 1$ indicates fraud and $y = 0$ indicates a legitimate transaction.

Although the core problem is binary classification, the project goes beyond a fixed-threshold setting by examining how classification decisions change when the threshold is tuned using validation data and when different business costs are assigned to false positives and false negatives.

4. Proposed Methods and Model Comparison Plan

To satisfy the supervised learning focus of the course, the following models will be implemented and compared:

1. **Logistic Regression**
   This will serve as a baseline linear classifier. It is simple, interpretable, and suitable for establishing a strong baseline. L1 and/or L2 regularization will be explored.
2. **Random Forest**
   This model can capture nonlinear feature interactions and is robust for tabular data. It also provides feature importance estimates that can be used for analysis.
3. **Support Vector Machine (SVM) or Gradient Boosted Trees**
   Depending on computational feasibility, either an SVM or a boosted tree model will be used as a third comparison model. This will allow comparison between margin-based learning and ensemble-based nonlinear learning.

In addition to standard model comparison, the project will evaluate:

- Class weighting
- Random under-sampling / over-sampling
- SMOTE (if appropriate)
- Threshold tuning using validation data
- Cost-sensitive evaluation under different error-cost assumptions

5. Planned Data Preprocessing

The planned preprocessing pipeline includes:

- Exploratory data analysis to understand the feature distributions and imbalance severity
- Inspection for missing values or anomalies

- Feature scaling for models that depend on feature magnitude, especially Logistic Regression and SVM
- Train/validation/test split to ensure fair and reproducible evaluation
- Possible rebalancing techniques applied only to the training set to avoid data leakage

Special attention will be paid to avoiding leakage and ensuring that any sampling method is performed only after the dataset is split.

6. Evaluation Metrics and Experimental Plan

Because the dataset is highly imbalanced, standard accuracy alone is not sufficient. The following metrics will be used:

- Precision
- Recall
- F1-score
- ROC-AUC
- Precision-Recall AUC
- Confusion Matrix

The experiments will compare:

- Baseline model performance
- Performance after imbalance-handling strategies
- Performance under different thresholds
- Precision-recall trade-offs under different cost assumptions

Hyperparameter tuning will be performed using validation data or cross-validation, and the final report will include comparisons of training and validation behavior, model robustness, and error analysis.

7. Novel Aspect of the Project

The key novelty of this project is that it does not stop at standard fraud classification. Instead, it studies the practical decision layer on top of prediction scores. Most student projects use a fixed threshold of 0.5, but this project will investigate how the threshold should change when the cost of missing fraud differs from the cost of incorrectly flagging legitimate transactions. This makes the project more realistic and more insightful than a standard benchmark comparison.

8. Expected Outcome

The expected outcome is a reproducible fraud detection pipeline that:

- compares multiple supervised learning models,
- demonstrates the effect of class imbalance on performance,
- shows how rebalancing and regularization affect detection quality,
- identifies the best-performing model under standard metrics,
- and analyzes how dynamic thresholding changes the practical usefulness of the system.

The project is expected to show that the best model is not necessarily the one with the highest accuracy, but rather the one that best balances fraud recall, false-positive rate, and decision cost.

9. Tools and Implementation

The project will be implemented in `Python` using:

- `NumPy` and `Pandas` for data processing
- `Matplotlib` and/or `Seaborn` for visualization
- `scikit-learn` for preprocessing, training, tuning, and evaluation
- `imbalanced-learn` for imbalance-handling methods such as SMOTE
- Jupyter Notebooks for experimentation and reproducibility

10. Deliverables

The final project is expected to produce:

- a complete machine learning pipeline,
- model comparison tables,
- threshold analysis plots,
- confusion matrices and performance curves,
- a reproducible GitHub repository with code and README,
- and a final report discussing methodology, experiments, results, limitations, and future work.